

1. DATA INTEGRITY REPORT - TASK 2

1. Missing Values

- **What I Found:** Checked all columns for missing values. 'return_date' had empty cells for ongoing checkouts. Other text columns had minor gaps.
- **Where It Was:** 'return_date', 'author', and 'genre' columns.
- **How Big It Was:** 'return_date' missing values matched unreturned books.
- **What I Did & Why:** Preserved 'return_date' missing values as NA because it reflects real logic (unreturned books). Filled missing metadata with 'Unknown' to preserve borrowing history.

2. Duplicate Records

- **What I Found:** Checked for exact identical rows across the dataset.
- **Where It Was:** Entire rows across combined columns.
- **How Big It Was:** A few exact duplicate rows from merging web data with database logs.
- **What I Did & Why:** Used drop_duplicates() to remove exact repeated rows. Kept similar rows with different checkout dates/IDs as real separate events.

3. Text Formatting and Inconsistencies

- **What I Found:** Categorical columns contained trailing spaces and mixed casing (e.g., 'heliopolis' vs 'Heliopolis ').
- **Where It Was:** 'neighborhood' and 'member_status' columns.
- **How Big It Was:** Multiple entries had spacing and capitalization differences.
- **What I Did & Why:** Used str.strip() for extra spaces and standardized capitalization (.str.title() for neighborhoods and .str.lower() for status) to aggregate categories correctly.

4. Invalid Member IDs

- **What I Found:** Checkout records referencing a 'member_id' not present in the registered members table.
- **Where It Was:** 'member_id' column in combined checkouts.
- **How Big It Was:** A small set of orphan checkout records.
- **What I Did & Why:** Filtered out orphan records using valid member IDs to ensure member statistics stay accurate without changing the official members list